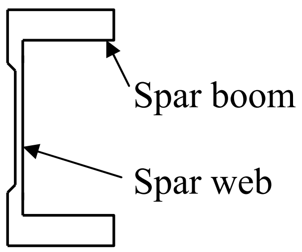
IDEALISATION OF SECTIONS

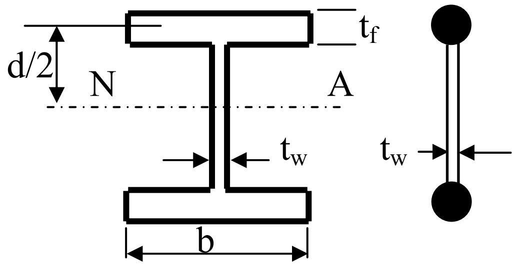
The components of an actual aircraft's structure may be extremely complex. To enable such structures to be analysed simplifying assumptions must be made , the number and nature of which determine the accuracy and degree of complexity of the analysis.

Spars



Here the web is considered to carry only the shear and the booms carry only end load.

Consider a spar section constructed from an I section beam.



The I section has the dimensions shown in the diagram. From the assumption above the flanges would be idealised as end load carrying areas and the web as a shear panel of thickness t_W.

When making such assumptions it is important that the basic properties of the section are not altered, here the boom and web have had their respective areas and thicknesses maintained, but how will this have affected the second moment of area?

In the original section:-

$$\begin{aligned} I_{NA} &= I_{FLANGES} + I_{WEB} \\ &= I_{CC} + A_f (d/2)^2 + I_{WEB} \end{aligned}$$

where I_{CC} = local second moment of area of flanges

A_f = area of flanges

$$\therefore I_{NA} = 2\left(\underbrace{\frac{bt_f^{\,3}}{12}}_{\text{Small – neglect}} + \underbrace{bt_f\left(\frac{d}{2}\right)^2}_{\text{large}} \right) + \underbrace{\frac{t_w d^3}{12}}_{\text{neglect}} = 2bt_f\left(\frac{d}{2}\right)^2 = \underline{\underline{2A_f\left(\frac{d}{2}\right)^2}}$$

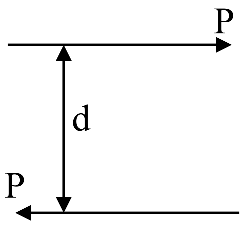
So long as the web thickness is small, the web and local second moment of the areas of the flanges will be small and only the $A(d/2)^2$ term of the flanges need be considered.

Consider stresses in the spar.

$$\sigma = \frac{My}{I} \quad \text{but} \quad I = 2A_f\left(\frac{d}{2}\right)^2$$

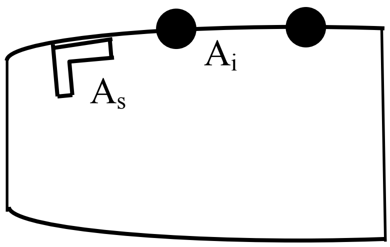
substituting we get:-

$$\sigma = \frac{M\left(d/2\right)}{2A_f\left(d/2\right)^2} = \frac{M}{dA_f}$$



now $M = P \times d$ therefore stress $= \dfrac{P}{A_f}$

Skin/Stiffener

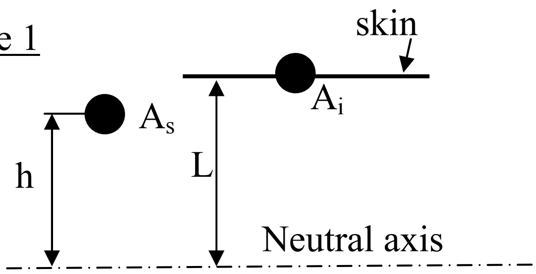


The spar only forms the vertical members of the complete wing box, the skin and stiffeners making up the remaining top and bottom surfaces, here the stringer (stiffener) is is often idealised at the skin line.

$A_i = A_s \times$ idealisation factor.

Here the stiffener has to carry the end load stresses in a similar manner to the spar booms. However, if the area of the stiffener is used at the skin line, this will over (or under in some cases) estimate the total second moment of area of the box, and if used in bending calculations would give stress lower than actually present. Therefore, in order to obtain the true box second moment of area, an idealisation factor must be applied to the stiffener area. Using the symbols shown in Figure 1 then:-

<u>Figure 1</u>



$$I_{centroid} = A_s \times h^2$$

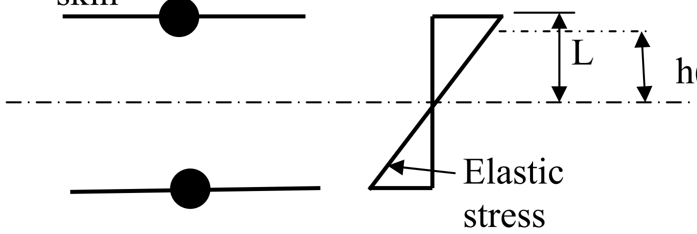
$$I_{idealised} = A_i \times L^2$$

$$A_s h^2 = A_i L^2 \quad \therefore A_i = A_s \frac{h^2}{L^2}$$

where $\dfrac{h^2}{L^2}$ = the idealisation factor

Again local second moment of area of the stiffener can be ignored.
The stresses yielded will now be correct at the skin line, but for an elastic analysis will be an over estimate if taken for the stiffener centroid or skin/stiffener stress, and should be de-idealised for these locations.

Figure 2



De-idealisation factor = h/L

Example 1

A box beam, symmetric about the X-X axis, and as shown in Figure 3 is to be idealised into four booms only, located at the box corners.

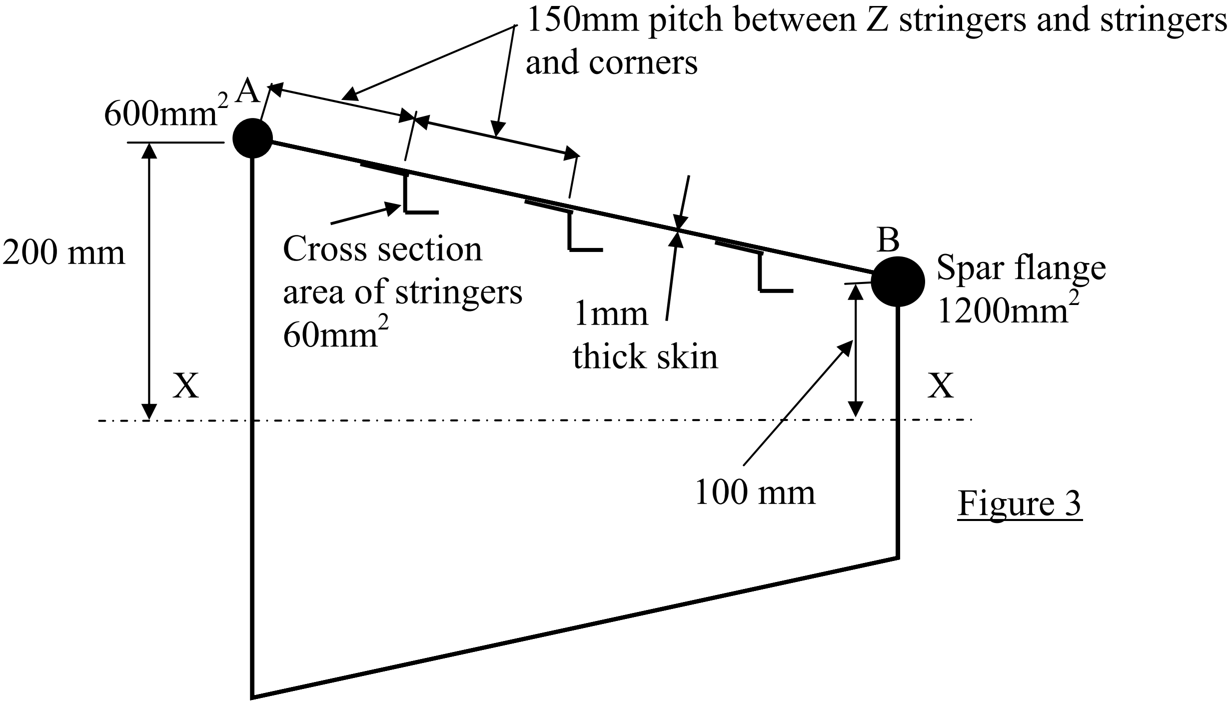
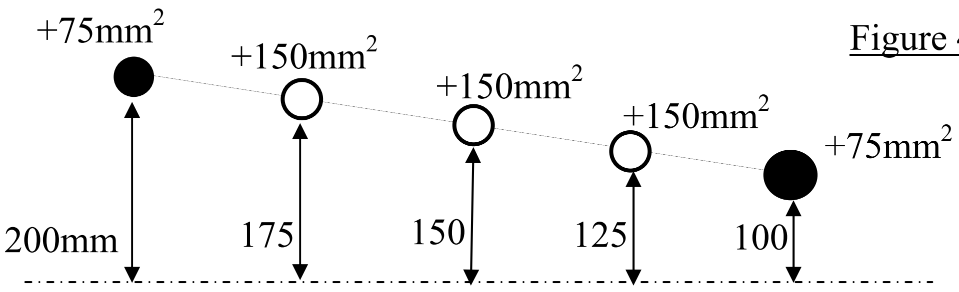


Figure 3

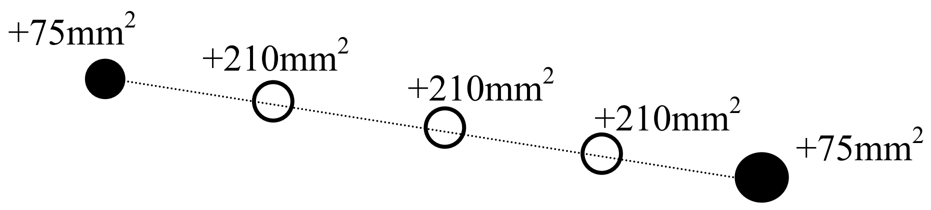
Addressing the top half of the box, first we apportion the skin at the stringer and spar locations as shown in Figure 4 .

Figure 4



By adding together the stringer areas and skin areas the total area at the stringer positions is obtained as shown in Figure 5.

Figure 5



These areas are then idealised at the corners A and B as follows:-

$$A_A \times 200^2 = 210 \times 175^2 + 105 \times 150^2$$

$$\underline{A_A = 220\text{mm}^2}$$

$$A_B \times 100^2 = 210 \times 125^2 + 105 \times 150^2$$

$$\underline{A_B = 564\text{mm}^2}$$

therefore the total boom area (A_A) at A is 220+675 = **895mm^2**

total boom area (A_B) at B is 564 + 1275 = **1839mm^2**

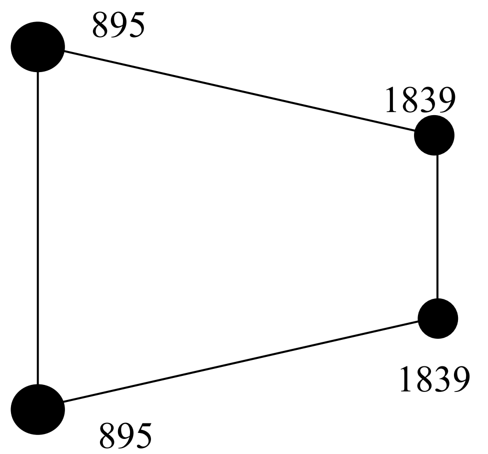
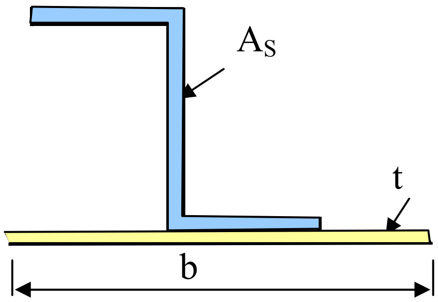


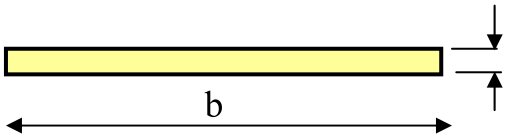
Figure 6
Total idealised boom areas

See tutorial question 1

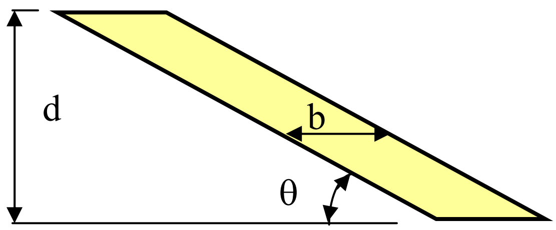
Alternative Method



For one stringer and skin pitch (b) this is equivalent to

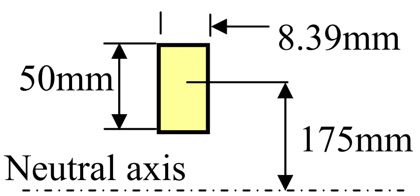


$$bt_E = A_S + bt \qquad \therefore\ t_E = t + \frac{A_S}{b} \qquad t_E = 1 + \frac{60}{150} = 1.4\text{mm}$$



For a parallelogram shape $I = bd^3/12$ where b is the "slant" width of the skin.

$$\sin\theta = 100/600 \quad \theta = 9.6^\circ \quad \therefore\ b = \frac{t_E}{\sin 9.6} = 8.39\text{mm}$$ .Dividing the skin into two then



$$A_A \times 200^2 = \frac{8.39 \times 50^3}{12} + (8.39 \times 50)175^2$$

$$A_A = \underline{323\text{mm}^2}$$

$$\text{Total } A_A = 323 + 600 = \underline{923\text{mm}^2} \quad \text{re} \quad 895\text{mm}^2$$

$$A_B \times 100^2 = \frac{8.39 \times 50^3}{12} + (8.39 \times 50)125^2$$

$$\underline{A_B = 664\text{mm}^2}$$

$$\underline{\underline{\text{Total } A_B = 664 + 1200 = 1864\text{ mm}^2}} \text{ re } 1839\text{mm}^2$$

Tutorial

Question 1

The symmetric cross section of a uniform single cell box is given in Figure Q1, the top and bottom skins are reinforced by identical stiffeners of $75mm^2$ cross section equally pitched, which together with the sheet are fully effective at carrying direct stresses. The vertical webs are effective at carrying only shear stresses.
Idealise the section into a four boom box, in which the sheet material carries only shear stresses, the end load carrying effect of the stiffeners and top and bottom skins being allowed for by the addition of effective area to the existing corner booms.
(N.B. Locate skin at stringers without idealising.)

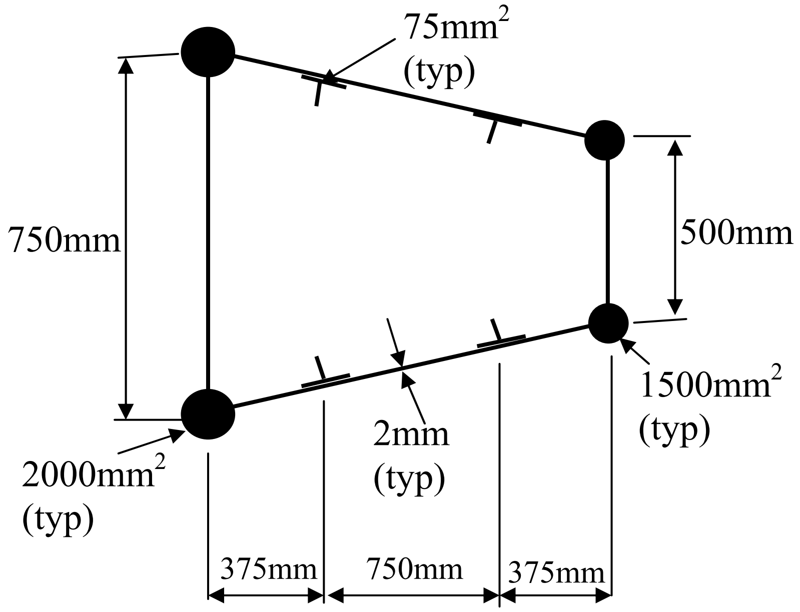


Figure Q1